A total of 10 LFR test deployments were conducted by the MPS between 2016 and 2019. An additional non-operational trial was conducted in April 2019. The sequence of test deployments was as follows:

- Notting Hill Carnival, 28-29 August 2016
- Notting Hill Carnival, 27-28 August 2017
- Remembrance Sunday Commemorations, Whitehall, 12 November 2017
- Port of Hull docks, assisting Humberside Police, 13-14 June 2018
- Stratford (Westfield), London, 28 June 2018
- Stratford (Westfield), London, 26 July 2018
- Soho (Cambridge Circus and Leicester Square), 17 December 2018
- Soho (Leicester Square), 18 December 2018
- Romford (Town Centre), 31 January 2019
- Romford (Town Centre), 14 February 2019.

The authors of this report observed the final six test deployments, beginning with the first deployment in Stratford (Westfield) in June 2018. Researchers did not observe a non-operational trial conducted in April 2019.

This report highlights a number of issues arising from the LFR test deployments. The report acknowledges the difficulties faced when seeking appropriate use of emerging technologies for the pursuit of public safety in a context where national guidance and leadership is largely absent. The report also notes a number of refinements to the approach adopted by the MPS as the test deployments progressed. However, the report raises some significant concerns. These are most notable with respect to: (a) the research process adopted by the MPS to trial LFR technology, and whether this was capable of addressing the identified objectives; (b) the absence of an explicit legal basis for the use of LFR, and concerns that the implicit legal basis identified by the MPS is inadequate in relation to the 'in accordance with the law' requirement established by human rights law; (c) the absence of effective analysis addressing the 'necessity in a democratic society' of the use of LFR in the circumstances trialled by the MPS, as required by human rights law; and (d) operational factors relating to inconsistency in the adjudication process, a presumption to intervene, problems with how the MPS engaged with individuals, and difficulties in obtaining the consent of those affected.

As detailed in this report, it is highly possible that the LFR trial process adopted by the MPS would be held unlawful if challenged before the courts.[6] This demonstrates

[6] It is noted that, at the time of writing, South Wales Police's trial of LFR is currently being challenged before the courts, indicating the existence of a case to answer with respect to the legality of certain elements of the LFR trial process. Further demonstrating the level of formal scrutiny being applied to this technology, the US House of Representatives Committee on Oversight and Reform held a hearing to examine the use of facial recognition technology by government and commercial entities, its impact on civil rights and liberties, and the need for oversight during the same week (https://oversight.house.gov/legislation/hearings/facial-recognition-technology-part-1-its-impact-on-our-civil-rights-and).

- 'Verified correct matches' are instances when a computer generated match is judged to be credible by a human operator and proven correct following an identity check.

'Watchlist' refers to the database of individuals against which live camera images were matched.[228]

4.1.2. Top Level Summary of Data

The MPS conducted 10 test deployments of LFR technology in London between 2016 and 2019. The authors became involved in the process part-way through and observed the final six of these. The test deployments were:

Table 4.1. MPS LFR Deployments Occurring Before Research commenced[229]

Trial	Date	Number of LFR matches	Watchlist size
Notting Hill Carnival 2016	28-29/08/2016	1	266
Notting Hill Carnival 2017	27-28/08/2017	96	528
Remembrance Sunday 2017	12/11/2017	7	42
Humberside Port 2018	13-14/06/2018	0	144

Table 4.2. Observed MPS LFR Deployments

Trial	Date	Number of LFR matches	Watchlist size
Stratford 1	28/6/2018	5	489
Stratford 2	26/7/2018	1	306
Soho 1	17/12/2018	5	2226
Soho 2	18/12/2018	9	2226
Romford 1	31/1/2019	10	2401
Romford 2	14/2/2019	16	1996

Overall, the LFR system generated 46 matches over the course of observed test deployments, involving 45 separate individuals.[230] Three of those matches are discounted from the analysis because they occurred before the street based intervention teams were deployed and, therefore, no capacity existed to attend to any

[228] In the literature on facial recognition, an image on the watchlist (reference database) is sometimes called a 'gallery image' while the imported image (in this instance, from the live camera feed) is called the 'probe image'.

[229] These test deployments were not observed by the research team; data was provided MPS facial recognition technical evaluation team.

[230] At the first Romford trial the same individual was matched twice at separate times in the same afternoon.

While not observed by the research team, and therefore only included for illustrative purposes, data from the alerts generated at the first four test deployments further demonstrate the problems with drawing definitive conclusions from comparing watchlist size and number of alerts. This data is also not comparable with that presented in the previous table given the use of a different, and apparently less effective, face recognition algorithm during the first three deployments. The intention here is to illustrate the wide and unreliable variance in the ratio between alerts and watchlist size.

Table 4.6. All LFR Alerts and Watchlist Sizes (First Four Trial Deployments)[234]

Trial	Date	Number of computer generated LFR matches	Watchlist size	Ratio of LFR matches to watchlist size
Notting Hill Carnival 2016	28-29/08/2016	1	266	1:266.00
Notting Hill Carnival 2017	27-28/08/2017	96	528	1:5.50
Remembrance Sunday 2017	12/11/2017	7	42	1:6.00
Humberside Port 2018	13-14/06/2018	0	144	n/a due to n=0

Taken together, the evidence presented by the test deployments demonstrate the shortcomings of any conclusions over LFR effectiveness based on counting absolute numbers of matches. It is also important to recognise how such calculations, outcomes and ratios are influenced by a range of additional and unaccounted for variables. These include the time in which the cameras were active and the density of crowds passing the cameras. The MPS technological evaluation of the deployments has reportedly developed a methodology in this regard, accounting for and weighting the influence of many such variables.

Determining false positives

Dispute exists over the most accurate way to measure true and false positives in respect to LFR computational matches. This is reflected in the clashing statistical claims made about LFR. One widely cited claim is that 98% of police LFR matches made nationally were false positives.[235] This appears to have been calculated on the basis of comparing publicly available data on, and Freedom of Information requests for, the number of LFR computer-generated matches against numbers of ultimately true positives, that being the number of individuals whose identity was confirmed

[234] These trial deployments were not observed by the research team, data was provided by the MPS facial recognition technical evaluation team.

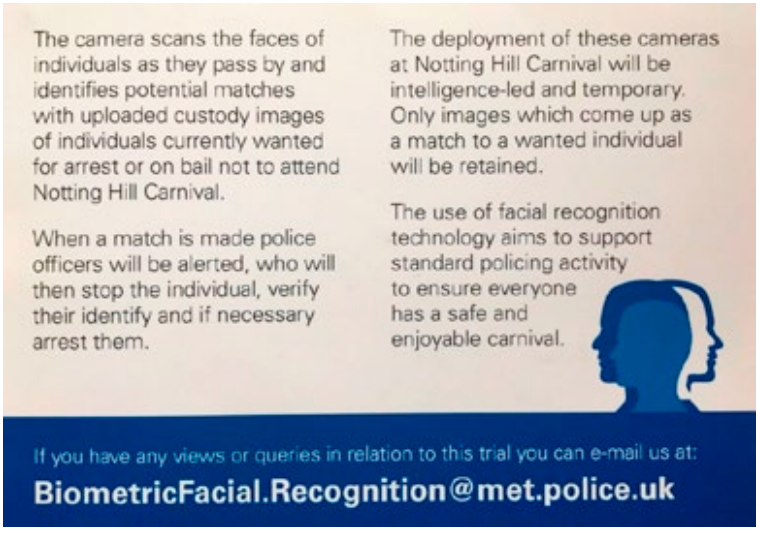
[235] Big Brother Watch, 'Face Off: The lawless growth of facial recognition in UK policing', 2018, p. 3.

Of additional note here is that leaflets for the Stratford test deployments (June and July 2018) contained less information on the purpose and likely subjects of surveillance than those provided at the earlier 2017 test deployment at the Notting Hill Carnival (see **Plates 4.3. & 4.4.).**

Plate 4.3. Metropolitan Police Service LFR Public Information Leaflet (Notting Hill Carnival test deployment, 2017, front)



Plate 4.4. Metropolitan Police Service LFR Public Information Leaflet (Notting Hill Carnival test deployment, 2017, back)



Some of the more detailed information was again included in updated leaflets used for the two Soho test deployments (**Plate 4.5**.). However, the less detailed Stratford leaflets were again used for the later Romford test deployments

For the first Soho trial [17 December 2018] the LFR van was stationed in Cambridge Circus, at a busy intersection between Shaftesbury Avenue and Charing Cross Road. As visible in **Plate 4.7.**, individuals would have already passed through the camera's field of view before being alerted to their presence by the signs. The LFR van used as a control room was also unmarked, preventing easy identification. Both of these issues were rectified to some degree when the van relocated to Leicester Square for the afternoon where an LFR sign was placed against the van, (see **Plate 4.9.**). However, there was no opportunity for an individual to exercise informed consent during the morning deployment.

Information boards were positioned further away from and outside of the cameras' field of view for each of the other test deployments. Questions exist over the degree to which this provided the detail and time needed for reflection in order to offer meaningful consent. Some signs placed close to the ground, as in both Soho trials, may be unnoticed as people pass by. This point was expressed by Liberty during their interview for this report: 'very small signs that people can't read isn't informing the public'.[286] Moreover, the high degree of visual and audible distractions experienced by pedestrians in some locations, such as Leicester Square (see **Plate 4.8.**), may further reduce the prominence of LFR signage. In all cases, an individual would have entered the cameras'[287] field of view within a few seconds of encountering an information board.

Other civil society organisations have also challenged the extent to which consent is informed or meaningful. Director of Big Brother Watch, Silkie Carlo, focused especially on the data processing and biometric processing aspects:

> No, there's no consent. No, there's no information, so people aren't informed. They're not in any meaningful sense of the word, even the vernacular word or the legal meaning of the word, there's no meaningful consent process whatsoever. You certainly can't withdraw consent. It has real life significance and legal impacts as well, serious risks to individuals involved. Data collection that people aren't informed about. And biometric data processing. … We've [probably spoken to] more people than anyone, members of the public, than any other group because we've been at every deployment except the first one in Stratford, we've been at every deployment since the announcement of the trial, when we were in Notting Hill Carnival, the second one, and then Stratford, Romford, Leicester Square, all the rest of it, and we've been there beginning to end handing out leaflets, talking to people. I would

[286] Interview with Hannah Couchman, Advocacy and Policy Officer, Liberty.

[287] All observed test deployments used two cameras. For 'static' deployments, such as Stratford, they were situated in close proximity to each other (see **Plate 4.6.**). For mobile deployments (Soho and Romford), they were located on top of the same van (see **Plate 4.9.)**. The two cameras worked in tandem, serving to each supplement and widen the other's field of view.